

Bachelor's Thesis

Degree in Industrial Technologies Engineering (GETI)

Development of a software to simulate three-point bending tests in composite laminates using Classical Lamination Theory and First-Order Shear Deformation Theory.

REPORT

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Abstract

This Bachelor's Thesis presents the development of a MATLAB software to simulate the mechanical behaviour of laminates made from composite materials, applying Classical Lamination Theory (CLT) and First-Order Shear Deformation Theory (FSDT). The main objective is to predict the structural response of sandwich panels subjected to three-point bending tests, as this is the most relevant test for evaluating stiffness and strength properties in the Formula Student competition — an engineering competition in which university teams design and build a racing car that they ultimately take to international competitions.

The work addresses the analysis of composite laminates from two complementary perspectives. On one hand, the in-plane behaviour of each individual ply is studied: how it responds elastically according to fibre orientation and under what loading state failure occurs. On the other hand, the analysis is extended to the full laminate scale and, finally, to the sandwich configuration.

Two plate theories are employed. Classical Lamination Theory (CLT), which is valid when the panel is sufficiently thin, becomes inadequate for sandwich panels where the core, far less stiff than the faces, deforms appreciably in shear. To capture this effect, First-Order Shear Deformation Theory (FSDT) is used, which relaxes the kinematic constraints and thereby incorporates transverse shear deformation [2].

The software incorporates both theories, yielding a model capable of predicting deflection, stress distribution and the safety factor against failure under three-point loading, thereby facilitating the design and optimisation of these components.

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Glossary

CLT Classical Lamination Theory.

FSDT First-Order Shear Deformation Theory.

CFRP Carbon Fibre Reinforced Polymer.

FPF First Ply Failure.

Twill A type of woven fabric pattern characterised by a diagonal weave.

1 Preface

1.1 Motivation

The main motivation for this project arises from the need for a practical and accurate tool capable of predicting the mechanical behaviour of various laminates under different structural tests. Historically, the Formula Student team Bcn eMotorsport has lacked a reliable predictive resource, which has forced the team to rely heavily on prior experience and empirical validation. Since the fabrication of this type of specimen is a laborious, costly and meticulous process, purely experimental iteration represents a significant delay in the design and development phases of the racing car.

Although finite element method (FEM) simulations using commercial software have been attempted in the past, the results have often lacked the desired accuracy. This lack of precision usually stems from the complexity of the mathematical formulation underlying these so-called “black-box” tools, whose correct configuration requires a deep understanding of the physical model being solved.

Finally, this project does not only aim to create a functional calculation tool, but also serves an educational purpose. The study and breakdown of plate theory is intended to provide the team with a solid theoretical foundation, enabling its members to understand composite material behaviour more rigorously and substantially facilitating any future research or design work.

1.2 Origin of the project

This project was carried out following my involvement with the Bcn eMotorsport team, where I was a member of the “Chassis & Composites” section for two seasons. This department is responsible for structural components, many of which are manufactured from composite materials — more specifically, sandwich panels. One example is the monocoque, one of the main structural elements of the car. Over those years I gained a comprehensive understanding of composite materials, covering both manufacturing and design. This experience is the driving force behind my motivation to delve deeper into this field and develop a tool that can help future generations and foster the growth of knowledge within the section.

2 Introduction

2.1 Objectives

The final objective of this work is to develop a MATLAB code capable of simulating the mechanical behaviour of composite sandwich panels subjected to three-point bending tests, allowing the stiffness and strength of the panel to be predicted. To achieve this, the following specific objectives are set out:

- Implement the constitutive relations of Classical Lamination Theory for orthotropic laminates of arbitrary orientation, including the construction of the ABD matrix.
- Calculate the effective elastic properties of a multidirectional laminate (Young's moduli, shear modulus and Poisson's ratios).
- Evaluate the strength of each ply in the laminate under a given loading state using the Tsai-Wu failure criterion, and obtain the global safety factor.
- Extend the model to one compatible with the sandwich panel, incorporating First-Order Shear Deformation Theory.
- Find an analytical solution for the three-point bending test and predict the maximum deflection, stress distribution and critical failure mode.

2.2 Project scope

The scope of this work comprises the computational development of a MATLAB tool for the study of composite laminates, bounded by the following considerations:

- **Material regime:** The analysis is restricted to the linear elastic behaviour of materials. Isothermal conditions are assumed, so thermal effects (such as curing residual stresses) and piezoelectric coupling phenomena are considered negligible. The focus is exclusively on mechanically induced loading.
- **Failure criterion:** First ply failure (FPF) analysis is implemented using classical failure criteria, without addressing progressive damage models or post-failure stiffness degradation.
- **Case study:** The simulation specialises in the three-point bending test. This choice responds to the Formula Student competition requirements, which establish this procedure as the official method for validating monocoque stiffness and strength.
- **Theoretical basis:** The mathematical formulation of the software implements Classical Lamination Theory (CLT) and First-Order Shear Deformation Theory (FSDT), based on the reference works of *Daniel & Ishai* [1] and *Reddy* [2].

3 Composite Materials

3.1 Definition and applications

A composite material is defined as a combination of two or more phases distinguishable at the macroscopic scale that act synergistically to achieve better properties than those of their individual components acting separately. In general, one phase is discontinuous, stiffer and stronger — known as the reinforcement — while the less stiff and less strong phase is continuous and is called the matrix. Occasionally, due to chemical interactions or processing effects, an additional distinct phase exists between the reinforcement and the matrix, called the interphase.

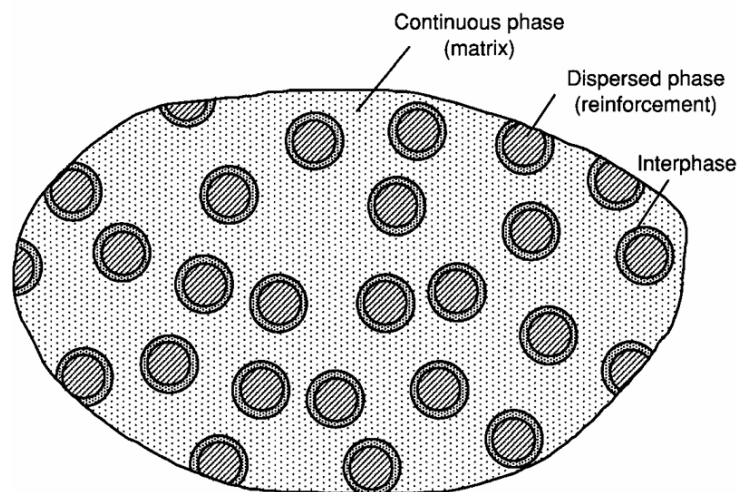
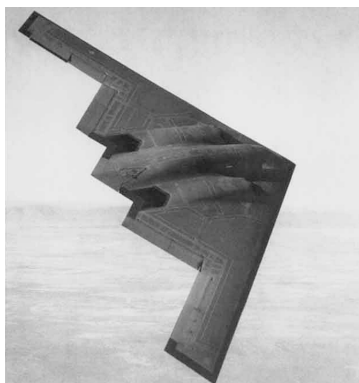


Figure 1: Phases of a composite material.

Composite materials have become a widely adopted option in aeronautical, automotive and naval engineering, thanks to their exceptional stiffness-strength-to-weight ratio. Compared to traditional structural materials such as aluminium or steel, they offer much higher specific moduli and tensile strengths, at the cost of greater expense and more complex handling, owing to their anisotropic nature.



(a) B-2 Stealth bomber, manufactured almost entirely from composite materials.



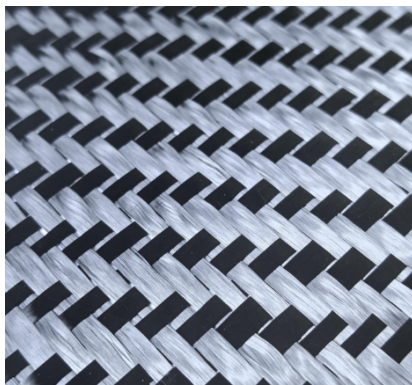
(b) Visby-class corvette of the Royal Swedish Navy, built with materials such as carbon/vinylester and PVC foam.

Figure 2: Examples of composite material applications.

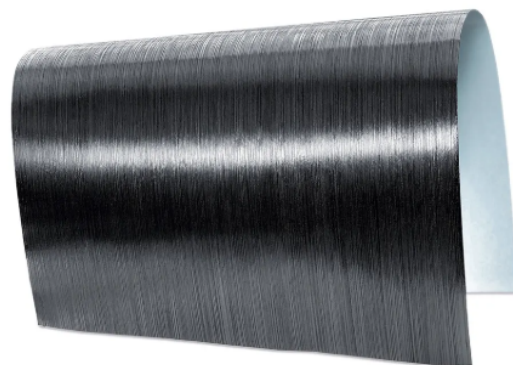
Weave patterns

In addition to the fibre type, the properties of a ply depend on the pattern in which the fibres are organised within the fabric. The two most commonly used weave types are:

- **Woven fibres:** fibres run in two perpendicular directions, interlacing to form a fabric. This provides a good balance of stiffness in both in-plane directions and good conformability over curved geometries. There are different types of interlacing; among the most popular is the *Twill* pattern.
- **Unidirectional fibres:** all fibres are oriented in a single direction. This offers maximum stiffness and strength in the fibre direction, but is practically inert in the transverse direction. It is used when the primary loading direction is well defined and targeted reinforcement is desired.



(a) *Twill* weave pattern.



(b) Unidirectional weave pattern.

Figure 3: Different weave patterns.

3.2 Sandwich Panel

Sandwich construction is of particular interest and is widely used because it is well-suited to lightweight structures with high in-plane and bending stiffness. A sandwich panel is a special type of laminate that typically consists of two thin face sheets and a lightweight, thicker, low-stiffness core that separates these faces, significantly increasing the section's moment of inertia — and therefore the bending stiffness — with a very small weight penalty. One of the most commonly used core geometries in structural applications is the *honeycomb*, whose cells follow a hexagonal pattern inspired by the natural honeycomb of bees.

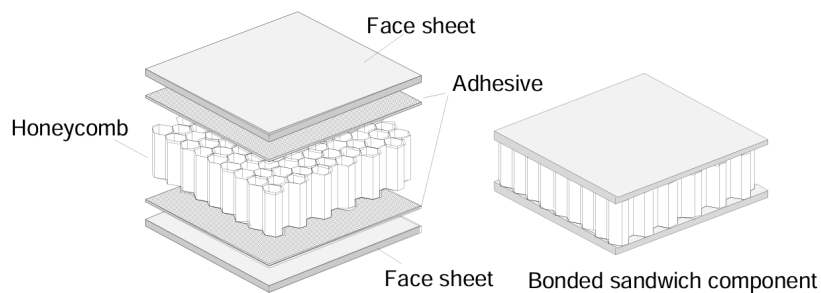


Figure 4: Elements of a sandwich panel.

The face sheets carry almost all of the bending and in-plane loads, while the core helps stabilise the faces against buckling and governs both the bending stiffness and the out-of-plane compressive and shear behaviour.

The hexagonal pattern of the *honeycomb* ensures a very low density while offering good mechanical properties under compressive and shear loading. The most relevant core properties are its density and its shear moduli G_{31} and G_{32} , which correspond to shear in the two transverse planes of the panel.

The most common structural cores are made of aluminium, Kevlar (aramid) or Nomex. For competition applications, aluminium or aramid cores are preferred, as they offer the best shear modulus-to-density ratio.

4 Theoretical foundations and implementation

This section presents in detail the mathematical foundations of Classical Lamination Theory as implemented in the MATLAB code. The exposition follows an ascending order of scale: first, the elastic response and strength of an individual ply are addressed, and then the results are extended to a multidirectional laminate composed of several plies.

4.1 Voigt notation

In composite mechanics it is common to work with the second-order stress tensor σ_{ij} , whose nine components represent the normal and shear stresses acting on the faces of a volume element, as illustrated in Figure 5.

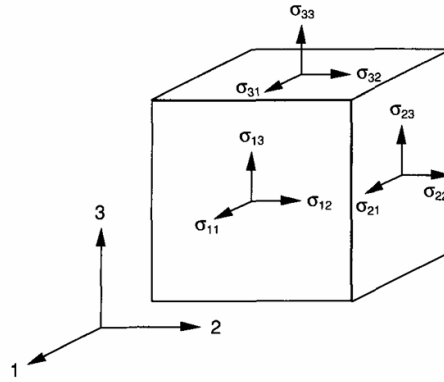


Figure 5: Components of the stress tensor on a volume element in the $(1, 2, 3)$ axis system.

To simplify the writing of constitutive relations, Voigt contracted notation is used, which reduces double subscripts to a single index according to the following convention:

$$\begin{aligned} \sigma_{11} &= \sigma_1, & \sigma_{22} &= \sigma_2, & \sigma_{33} &= \sigma_3, \\ \sigma_{23} &= \tau_{23} = \sigma_4 = \tau_4, & \sigma_{31} &= \tau_{31} = \sigma_5 = \tau_5, & \sigma_{12} &= \tau_{12} = \sigma_6 = \tau_6, \end{aligned} \quad (1)$$

and for strains:

$$\begin{aligned} \varepsilon_{11} &= \varepsilon_1, & \varepsilon_{22} &= \varepsilon_2, & \varepsilon_{33} &= \varepsilon_3, \\ 2\varepsilon_{23} &= \gamma_{23} = \varepsilon_4 = \gamma_4, & 2\varepsilon_{31} &= \gamma_{31} = \varepsilon_5 = \gamma_5, & 2\varepsilon_{12} &= \gamma_{12} = \varepsilon_6 = \gamma_6. \end{aligned} \quad (2)$$

Note that engineering shear strains (γ_{ij}) are twice the strain tensor components (ε_{ij}). Thanks to this notation, the fourth-order stiffness tensor C_{ijkl} and compliance tensor S_{ijkl} reduce to 6×6 matrices, C_{IJ} and S_{IJ} , and the constitutive relations can be written compactly as $[\sigma] = [C][\varepsilon]$ and $[\varepsilon] = [S][\sigma]$.

4.2 Ply and laminate: concepts and notation

A **ply** (*lamina*) is a flat layer of composite material formed by unidirectional or woven fibres embedded in a matrix. In the case of unidirectional fibres, one speaks of a *UD ply*; in the case of a woven fabric, the warp and fill directions define the two principal in-plane axes. The ply is the basic element of the analysis: it is an orthotropic material with three planes of material symmetry, and its principal axes are designated 1, 2 and 3, where axis 1 is the longitudinal fibre direction (or warp direction), axis 2 is the transverse direction in the ply plane, and axis 3 is normal to the ply plane [1].

A **laminate** consists of two or more plies stacked together, usually at different orientations. Since the orientation of the principal axes varies from ply to ply, it is more convenient to analyse the assembly in a common reference frame (x, y, z) fixed to the laminate, as illustrated in Figure 6. The orientation of each ply is defined by the angle θ measured counter-clockwise from the laminate x -axis to the ply's axis 1.

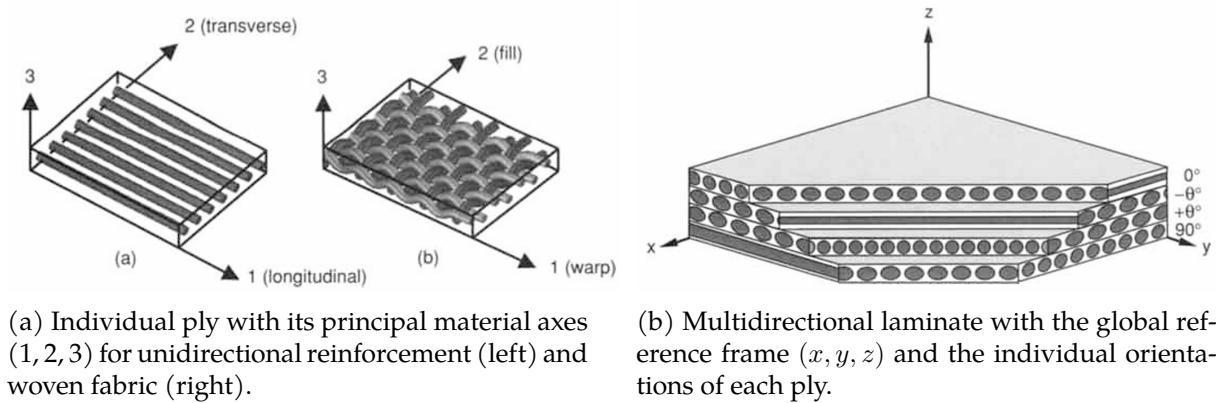


Figure 6: Definition of ply and laminate with their reference frames [1].

The configuration of a laminate is described by the **stacking sequence**, which specifies the number of plies, the material of each one, its orientation and its stacking order. For example, $[0/ - 45/45/90]_s$ indicates four plies with those orientations; the subscript s indicates that the laminate is symmetric about the mid-plane, so the laminate actually has eight plies in total. This notation is the one used throughout this work to define the laminates analysed.

4.3 Elastic behaviour of a ply

Specialty orthotropic material

A ply of composite material — whether woven or unidirectional — is a specialty orthotropic material: it has three mutually perpendicular planes of material symmetry and the coordinate reference frame is selected along the principal planes of symmetry. In this case, the number of independent elastic constants reduces to nine, and the constitutive relations in Voigt notation take the form:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_4 \\ \tau_5 \\ \tau_6 \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{21} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{31} & C_{32} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_4 \\ \gamma_5 \\ \gamma_6 \end{bmatrix} \quad (3)$$

The inverse relation, which expresses strains as a function of stresses through the compliance matrix $[S]$, has the same structure:

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_4 \\ \gamma_5 \\ \gamma_6 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & 0 & 0 & 0 \\ S_{21} & S_{22} & S_{23} & 0 & 0 & 0 \\ S_{31} & S_{32} & S_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & S_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & S_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & S_{66} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_4 \\ \tau_5 \\ \tau_6 \end{bmatrix} \quad (4)$$

Plane stress assumption and derivation of the $[Q]$ matrix

In most structural applications, composite materials are used as thin plies loaded in their own plane. It can be assumed that they are in a state of **plane stress**, in which all out-of-plane stress components (direction 3) are zero:

$$\sigma_3 = 0, \quad \tau_{23} = \tau_4 = 0, \quad \tau_{13} = \tau_5 = 0. \quad (5)$$

Applying these conditions to Equation (3), the constitutive relations become:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ 0 \\ 0 \\ 0 \\ \tau_6 \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & C_{13} & 0 & 0 & 0 \\ C_{12} & C_{22} & C_{23} & 0 & 0 & 0 \\ C_{13} & C_{23} & C_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \varepsilon_3 \\ \gamma_4 \\ \gamma_5 \\ \gamma_6 \end{bmatrix} \quad (6)$$

Expanding the components yields the following system of equations:

$$\begin{aligned} \sigma_1 &= C_{11}\varepsilon_1 + C_{12}\varepsilon_2 + C_{13}\varepsilon_3 \\ \sigma_2 &= C_{12}\varepsilon_1 + C_{22}\varepsilon_2 + C_{23}\varepsilon_3 \\ 0 &= C_{13}\varepsilon_1 + C_{23}\varepsilon_2 + C_{33}\varepsilon_3 \\ \tau_6 &= C_{66}\gamma_6 \end{aligned} \quad (7)$$

The third equation allows ε_3 to be expressed as a function of ε_1 and ε_2 :

$$\varepsilon_3 = -\frac{C_{13}\varepsilon_1 + C_{23}\varepsilon_2}{C_{33}}. \quad (8)$$

Substituting (8) into the first two equations of (7) to eliminate ε_3 , the reduced stiffness relations are obtained:

$$\begin{aligned}\sigma_1 &= \left(C_{11} - \frac{C_{13}C_{13}}{C_{33}} \right) \varepsilon_1 + \left(C_{12} - \frac{C_{13}C_{23}}{C_{33}} \right) \varepsilon_2 = Q_{11}\varepsilon_1 + Q_{12}\varepsilon_2 \\ \sigma_2 &= \left(C_{12} - \frac{C_{23}C_{13}}{C_{33}} \right) \varepsilon_1 + \left(C_{22} - \frac{C_{23}C_{23}}{C_{33}} \right) \varepsilon_2 = Q_{12}\varepsilon_1 + Q_{22}\varepsilon_2 \\ \tau_6 &= C_{66}\gamma_6 = Q_{66}\gamma_6\end{aligned}\quad (9)$$

where the components of the **reduced stiffness matrix** $[Q]$ are defined in general as:

$$Q_{ij} = C_{ij} - \frac{C_{i3}C_{j3}}{C_{33}}, \quad (i, j = 1, 2, 6). \quad (10)$$

In compact matrix form, the constitutive relations under plane stress become:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix} = \underbrace{\begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}}_{[Q]} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_6 \end{bmatrix} \quad (11)$$

Their components are expressed in terms of the engineering constants E_1 , E_2 , G_{12} and ν_{12} as:

$$\begin{aligned}Q_{11} &= \frac{E_1}{1 - \nu_{12}\nu_{21}}, & Q_{22} &= \frac{E_2}{1 - \nu_{12}\nu_{21}}, \\ Q_{12} &= \frac{\nu_{21}E_1}{1 - \nu_{12}\nu_{21}} = \frac{\nu_{12}E_2}{1 - \nu_{12}\nu_{21}}, & Q_{66} &= G_{12},\end{aligned}\quad (12)$$

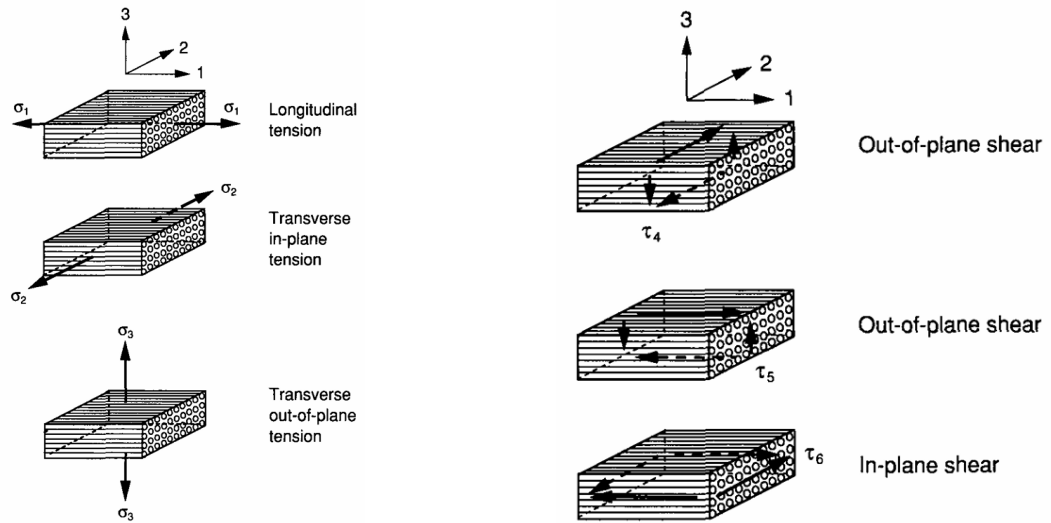
where the symmetry of the compliance matrix imposes the reciprocity relation:

$$\nu_{21} = \nu_{12} \frac{E_2}{E_1}. \quad (13)$$

The inverse relation is defined as:

$$\begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_6 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix} \quad (14)$$

The relations between the mathematical constants (Q_{ij}) and the engineering constants are established by means of elementary imaginary tests, as illustrated in the following figures. The derivation of these relations is not presented in this work, but can be found in the work of Daniel & Ishai [1].



(a) Elementary tests to obtain relations between mathematical and engineering constants (normal stress).

(b) Elementary tests to obtain relations between mathematical and engineering constants (shear stress).

Figure 7: Elementary tests to obtain relations between mathematical and engineering constants.

The MATLAB implementation of the $[Q]$ matrix is:

```

1 % Elastic properties of the ply
2 E1 = 150000; % MPa
3 E2 = 150000; % MPa
4 G12 = 5000; % MPa
5 nu12 = 0.3;
6
7 % Reciprocity relation
8 nu21 = E2 * nu12 / E1;
9
10 % Components of the reduced stiffness matrix
11 Q11 = E1 / (1 - nu12*nu21);
12 Q22 = E2 / (1 - nu12*nu21);
13 Q12 = (nu21*E1) / (1 - nu12*nu21);
14 Q66 = G12;
15
16 Q = [Q11 Q12 0;
17      Q12 Q22 0;
18      0 0 Q66]

```

Listing 1: Construction of the $[Q]$ matrix in principal axes

Coordinate transformation (2D)

In a real laminate, each ply may be oriented at an angle θ with respect to the laminate reference axis (x -axis). To transform stresses and strains between the ply's principal axes (1, 2) and the laminate axes (x, y), the **transformation matrix** $[T]$ is used:

$$[T] = \begin{bmatrix} m^2 & n^2 & 2mn \\ n^2 & m^2 & -2mn \\ -mn & mn & m^2 - n^2 \end{bmatrix}, \quad m = \cos \theta, \quad n = \sin \theta, \quad (15)$$

such that:

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix} = [T] \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_s \end{bmatrix}, \quad \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \frac{1}{2}\gamma_6 \end{bmatrix} = [T] \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \frac{1}{2}\gamma_s \end{bmatrix}. \quad (16)$$

Inverting the relations also gives:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_s \end{bmatrix} = [T^{-1}] \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix}, \quad \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \frac{1}{2}\gamma_s \end{bmatrix} = [T^{-1}] \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \frac{1}{2}\gamma_6 \end{bmatrix}. \quad (17)$$

where:

$$[T^{-1}] = [T(-\theta)] = \begin{bmatrix} m^2 & n^2 & -2mn \\ n^2 & m^2 & 2mn \\ mn & -mn & m^2 - n^2 \end{bmatrix} \quad (18)$$

The angle θ is measured positive counter-clockwise from the x -axis to the ply's axis 1, as shown in the following figure.

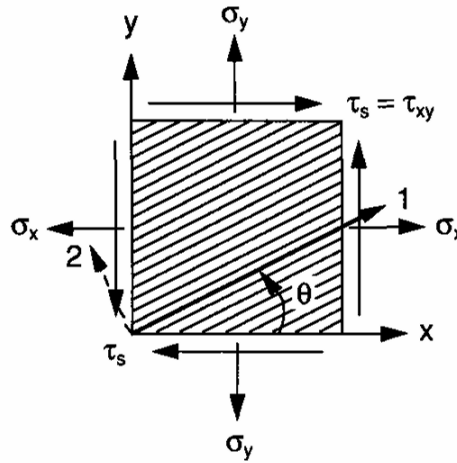


Figure 8: Relationship between the principal axes of the ply (1, 2) and the laminate reference axes (x, y) for an orientation angle θ .

Transformed stiffness matrix $[Q]_{x,y}$

When the ply is loaded along arbitrary axes x and y , the stress-strain relations expressed in those axes are:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_s \end{bmatrix} = \underbrace{\begin{bmatrix} Q_{xx} & Q_{xy} & Q_{xs} \\ Q_{yx} & Q_{yy} & Q_{ys} \\ Q_{sx} & Q_{sy} & Q_{ss} \end{bmatrix}}_{[Q]_{x,y}} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_s \end{bmatrix} \quad (19)$$

It is therefore necessary to know the relation between the transformed stiffness matrix $[Q]_{x,y}$ and the stiffness matrix in principal axes $[Q]_{1,2}$. Equation (19) can be rewritten as:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_s \end{bmatrix} = \begin{bmatrix} Q_{xx} & Q_{xy} & 2Q_{xs} \\ Q_{yx} & Q_{yy} & 2Q_{ys} \\ Q_{sx} & Q_{sy} & 2Q_{ss} \end{bmatrix} \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \frac{1}{2}\gamma_s \end{bmatrix} \quad (20)$$

Introducing the stress-strain relations of Equation (11) into Equation (16):

$$\begin{aligned} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_s \end{bmatrix} &= [T]^{-1} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix} = [T]^{-1} \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_6 \end{bmatrix} \\ &= [T]^{-1} \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & 2Q_{66} \end{bmatrix} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \frac{1}{2}\gamma_6 \end{bmatrix} = [T]^{-1} \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & 2Q_{66} \end{bmatrix} [T] \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \frac{1}{2}\gamma_s \end{bmatrix} \end{aligned} \quad (21)$$

Comparing the above equations leads to the transformation relation for the stiffness matrix:

$$\begin{bmatrix} Q_{xx} & Q_{xy} & 2Q_{xs} \\ Q_{yx} & Q_{yy} & 2Q_{ys} \\ Q_{sx} & Q_{sy} & 2Q_{ss} \end{bmatrix} = [T]^{-1} \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{21} & Q_{22} & 0 \\ 0 & 0 & 2Q_{66} \end{bmatrix} [T] \quad (4.66)$$

The explicit components of $[Q]_{x,y}$ in terms of m and n are:

$$\begin{aligned} Q_{xx} &= m^4 Q_{11} + n^4 Q_{22} + 2m^2 n^2 Q_{12} + 4m^2 n^2 Q_{66}, \\ Q_{yy} &= n^4 Q_{11} + m^4 Q_{22} + 2m^2 n^2 Q_{12} + 4m^2 n^2 Q_{66}, \\ Q_{xy} &= m^2 n^2 Q_{11} + m^2 n^2 Q_{22} + (m^4 + n^4) Q_{12} - 4m^2 n^2 Q_{66}, \\ Q_{xs} &= m^3 n Q_{11} - m n^3 Q_{22} - m n (m^2 - n^2) Q_{12} - 2m n (m^2 - n^2) Q_{66}, \\ Q_{ys} &= m n^3 Q_{11} - m^3 n Q_{22} + m n (m^2 - n^2) Q_{12} + 2m n (m^2 - n^2) Q_{66}, \\ Q_{ss} &= m^2 n^2 Q_{11} + m^2 n^2 Q_{22} - 2m^2 n^2 Q_{12} + (m^2 - n^2)^2 Q_{66}. \end{aligned} \quad (22)$$

```

1 theta = pi/4;    % angle in radians (positive counter-clockwise from x to
   axis 1)
2 m = cos(theta);
3 n = sin(theta);
4
5 T = [ m^2,    n^2,    2*m*n;

```

```

6      n^2,    m^2,    -2*m*n;
7      -m*n,    m*n,    m^2-n^2];
8
9      % The third column of the rewritten Q carries the factor 2 in Q66
10     Q_rewritten = [Q11    Q12    0;
11                   Q12    Q22    0;
12                   0      0    2*Q66];
13
14     % T \ Q_rewritten * T = inv(T) * Q_rewritten * T
15     Qxy = T \ Q_rewritten * T

```

Listing 2: Calculation of the transformed $[Q]_{x,y}$ matrix

Transformed engineering properties

The transformed compliance matrix $[\bar{S}]$ (inverse of $[\bar{Q}]$) can be expressed directly in terms of engineering constants referred to the x, y axes:

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_s \end{bmatrix} = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{yx}}{E_x} & \frac{\eta_{sx}}{G_{xy}} \\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & \frac{\eta_{sy}}{G_{xy}} \\ \frac{\eta_{xs}}{E_x} & \frac{\eta_{ys}}{E_y} & \frac{1}{G_{xy}} \end{bmatrix} \begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_s \end{bmatrix} \quad (23)$$

The explicit expressions for each term of $[\bar{S}]$ in terms of engineering constants and angle θ are:

$$\begin{aligned} \frac{1}{E_x} &= \frac{m^2}{E_1}(m^2 - n^2\nu_{12}) + \frac{n^2}{E_2}(n^2 - m^2\nu_{21}) + \frac{m^2n^2}{G_{12}}, \\ \frac{1}{E_y} &= \frac{n^2}{E_1}(n^2 - m^2\nu_{12}) + \frac{m^2}{E_2}(m^2 - n^2\nu_{21}) + \frac{m^2n^2}{G_{12}}, \\ \frac{1}{G_{xy}} &= \frac{4m^2n^2}{E_1}(1 + \nu_{12}) + \frac{4m^2n^2}{E_2}(1 + \nu_{21}) + \frac{(m^2 - n^2)^2}{G_{12}}. \end{aligned} \quad (24)$$

```

1 inv_Ex = m^2/E1*(m^2 - n^2*nu12) + n^2/E2*(n^2 - m^2*nu21) + m^2*n^2/
   G12;
2 inv_Ey = n^2/E1*(n^2 - m^2*nu12) + m^2/E2*(m^2 - n^2*nu21) + m^2*n^2/
   G12;
3 inv_Gxy = 4*m^2*n^2/E1*(1+nu12) + 4*m^2*n^2/E2*(1+nu21) + (m^2-n^2)^2/
   G12;
4
5 nuxy_Ex = m^2/E1*(m^2*nu12 - n^2) + n^2/E2*(n^2*nu21 - m^2) + m^2*n^2/
   G12;
6 etaxs_Ex = 2*m^3*n/E1*(1+nu12) - 2*m*n^3/E2*(1+nu21) - m*n*(m^2-n^2)/
   G12;
7 etays_Ey = 2*m*n^3/E1*(1+nu12) - 2*m^3*n/E2*(1+nu21) + m*n*(m^2-n^2)/
   G12;
8
9 S_xy = [inv_Ex, -nuxy_Ex, etaxs_Ex;

```

```

10      -nuxy_Ex,   inv_Ey,   etays_Ey;
11      etaxs_Ex,  etays_Ey,  inv_Gxy];
12
13  Ex_lamina = 1/S_xy(1,1)
14  Ey_lamina = 1/S_xy(2,2)

```

Listing 3: Transformed compliance matrix and engineering properties

Figure 9 summarises the complete calculation flow to obtain the elastic properties of an arbitrarily oriented ply. Starting from the engineering constants in principal axes ($E_1, E_2, G_{12}, \nu_{12}$), the mathematical matrices $[Q]_{1,2}$ and $[S]_{1,2}$ are constructed. Applying the transformation with the fibre orientation angle θ , the transformed matrices $[\bar{Q}]_{x,y}$ and $[\bar{S}]_{x,y}$ are obtained, from which the engineering constants referred to the laminate axes $[E]_{x,y}$ are finally extracted.

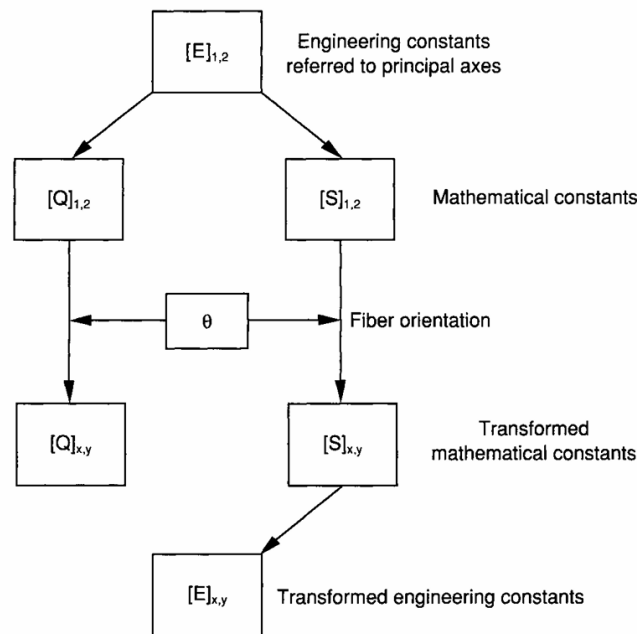


Figure 9: Schematic of the transformation process for the elastic properties of a ply from the principal axes to the laminate reference axes.

4.4 Ply strength: Tsai-Wu criterion

To predict failure of an orthotropic ply, the **Tsai-Wu criterion** is used, which accounts for the difference between tensile and compressive strengths. Under plane stress, the failure surface is expressed as:

$$f_1\sigma_1 + f_2\sigma_2 + f_{11}\sigma_1^2 + f_{22}\sigma_2^2 + f_{66}\tau_6^2 + 2f_{12}\sigma_1\sigma_2 = 1 \quad (25)$$

where the strength coefficients are defined from the uniaxial strengths $F_{1t}, F_{1c}, F_{2t}, F_{2c}$ and the shear strength F_6 as:

$$\begin{aligned}
f_1 &= \frac{1}{F_{1t}} - \frac{1}{F_{1c}}, & f_2 &= \frac{1}{F_{2t}} - \frac{1}{F_{2c}}, \\
f_{11} &= \frac{1}{F_{1t}F_{1c}}, & f_{22} &= \frac{1}{F_{2t}F_{2c}}, \\
f_{66} &= \frac{1}{F_6^2}, & f_{12} &= -\frac{1}{2}\sqrt{f_{11}f_{22}}.
\end{aligned} \tag{26}$$

The conservative estimate $f_{12} = -\frac{1}{2}\sqrt{f_{11}f_{22}}$ is recommended in the absence of biaxial test data.

Safety factor

To calculate how many times the applied load can be scaled before failure, the stress state is scaled by a factor k and the quadratic equation is solved:

$$a k^2 + b k - 1 = 0, \tag{27}$$

where $a = f_{11}\sigma_1^2 + f_{22}\sigma_2^2 + f_{66}\tau_6^2 + 2f_{12}\sigma_1\sigma_2$ and $b = f_1\sigma_1 + f_2\sigma_2$.

The two positive solutions correspond to the safety factor under forward loading (SF_a) and reverse loading (SF_r):

$$SF_a = \frac{-b + \sqrt{b^2 + 4a}}{2a}, \quad SF_r = \left| \frac{-b - \sqrt{b^2 + 4a}}{2a} \right|. \tag{28}$$

The global laminate safety factor is the minimum of the SF_a and SF_r values calculated at the top and bottom faces of each ply.

4.5 Elastic behaviour of multidirectional laminates

Green-Lagrange strain tensor

To quantify the deformation of a continuous solid it is necessary to compare the geometry of the body before and after loading. Let $\mathcal{C}^0$ be the **reference configuration** (undeformed state) and $\mathcal{C}$ the deformed configuration at time $t > 0$. In the **Lagrangian** description, positions are expressed through **material coordinates** (X_1, X_2, X_3): a particle occupying position $\mathbf{X}$ in $\mathcal{C}^0$ moves to position $\mathbf{x}$ in $\mathcal{C}$ (Figure 10). The **displacement vector** of the particle, in the Lagrangian description, is [2]:

$$u_i(X_1, X_2, X_3, t) = x_i(X_1, X_2, X_3, t) - X_i \tag{29}$$

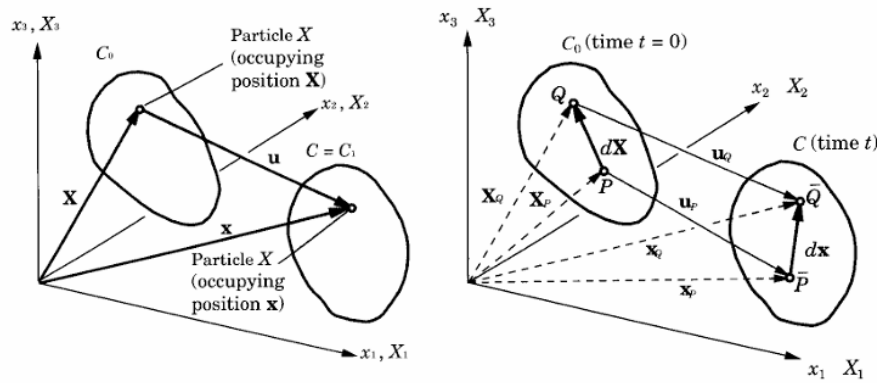


Figure 10: Kinematics of deformation of a continuum. (Left) Reference configuration C^0 and deformed configuration C . (Right) Position and displacement vectors of two neighbouring particles P and Q before and after deformation [2].

To measure deformation, the distances between two neighbouring particles P and Q are compared. In the reference configuration, the vector joining them is $d\mathbf{X}$, with length $dS = \sqrt{dX_i dX_i}$. After deformation, this vector becomes $d\mathbf{x}$, with length $ds = \sqrt{dx_i dx_i}$. The **Green-Lagrange strain tensor** $\mathbf{E}$ is defined from the change in the square of the length:

$$2 dX_i E_{ij} dX_j \equiv (ds)^2 - (dS)^2 = dx_i dx_i - dX_i dX_i \quad (30)$$

To express $\mathbf{E}$ in terms of the displacement field, we start from the position relation $\mathbf{x} = \mathbf{X} + \mathbf{u}(X_1, X_2, X_3, t)$, whose total differential, applying the chain rule, is:

$$d\mathbf{x} \equiv d\mathbf{X} + d\mathbf{X} \cdot \nabla \mathbf{u} = d\mathbf{X} \cdot (\mathbf{I} + \nabla \mathbf{u}) \quad (31)$$

where ∇ denotes the gradient operator with respect to material coordinates $\mathbf{X}$. Substituting (31) into definition (30):

$$\begin{aligned} 2 d\mathbf{X} \cdot \mathbf{E} \cdot d\mathbf{X} &= d\mathbf{x} \cdot d\mathbf{x} - d\mathbf{X} \cdot d\mathbf{X} \\ &= [d\mathbf{X} \cdot (\mathbf{I} + \nabla \mathbf{u})] \cdot [d\mathbf{X} \cdot (\mathbf{I} + \nabla \mathbf{u})] - d\mathbf{X} \cdot d\mathbf{X} \\ &= d\mathbf{X} \cdot (\mathbf{I} + \nabla \mathbf{u}) \cdot (\mathbf{I} + \nabla \mathbf{u})^T \cdot d\mathbf{X} - d\mathbf{X} \cdot d\mathbf{X} \\ &= d\mathbf{X} \cdot [(\mathbf{I} + \nabla \mathbf{u}) \cdot (\mathbf{I} + \nabla \mathbf{u})^T - \mathbf{I}] \cdot d\mathbf{X} \end{aligned} \quad (32)$$

from which the expression of the Green-Lagrange tensor in terms of the displacement gradient is obtained:

$$\mathbf{E} = \frac{1}{2} [(\mathbf{I} + \nabla \mathbf{u}) \cdot (\mathbf{I} + \nabla \mathbf{u})^T - \mathbf{I}] = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T + \nabla \mathbf{u} \cdot (\nabla \mathbf{u})^T] \quad (33)$$

Note that the tensor is symmetric, $\mathbf{E} = \mathbf{E}^T$. In rectangular Cartesian components, Equation (33) is equivalent to:

$$E_{jk} = \frac{1}{2} \left(\frac{\partial u_j}{\partial X_k} + \frac{\partial u_k}{\partial X_j} + \frac{\partial u_m}{\partial X_j} \frac{\partial u_m}{\partial X_k} \right) \quad (34)$$

The six independent components of tensor E_{jk} , written explicitly, are:

$$\begin{aligned} E_{11} &= \frac{\partial u_1}{\partial X_1} + \frac{1}{2} \left[\left(\frac{\partial u_1}{\partial X_1} \right)^2 + \left(\frac{\partial u_2}{\partial X_1} \right)^2 + \left(\frac{\partial u_3}{\partial X_1} \right)^2 \right] \\ E_{22} &= \frac{\partial u_2}{\partial X_2} + \frac{1}{2} \left[\left(\frac{\partial u_1}{\partial X_2} \right)^2 + \left(\frac{\partial u_2}{\partial X_2} \right)^2 + \left(\frac{\partial u_3}{\partial X_2} \right)^2 \right] \\ E_{33} &= \frac{\partial u_3}{\partial X_3} + \frac{1}{2} \left[\left(\frac{\partial u_1}{\partial X_3} \right)^2 + \left(\frac{\partial u_2}{\partial X_3} \right)^2 + \left(\frac{\partial u_3}{\partial X_3} \right)^2 \right] \\ E_{12} &= \frac{1}{2} \left(\frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} + \frac{\partial u_1}{\partial X_1} \frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} \frac{\partial u_2}{\partial X_2} + \frac{\partial u_3}{\partial X_1} \frac{\partial u_3}{\partial X_2} \right) \\ E_{13} &= \frac{1}{2} \left(\frac{\partial u_1}{\partial X_3} + \frac{\partial u_3}{\partial X_1} + \frac{\partial u_1}{\partial X_1} \frac{\partial u_1}{\partial X_3} + \frac{\partial u_2}{\partial X_1} \frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_1} \frac{\partial u_3}{\partial X_3} \right) \\ E_{23} &= \frac{1}{2} \left(\frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_2} + \frac{\partial u_1}{\partial X_2} \frac{\partial u_1}{\partial X_3} + \frac{\partial u_2}{\partial X_2} \frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_2} \frac{\partial u_3}{\partial X_3} \right) \end{aligned} \quad (35)$$

When displacement gradients are sufficiently small ($|\nabla \mathbf{u}| \ll 1$), the quadratic products in (34) are negligible compared to the linear terms, and the Green-Lagrange tensor reduces to the **infinitesimal strain tensor** ε :

$$\varepsilon = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T], \quad \varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (36)$$

This tensor is the strain measure adopted in CLT under the small deformation and small rotation assumption, detailed in the following section.

Hypotheses of Classical Lamination Theory

CLT is an extension of classical plate theory to composite laminates. Its formulation rests on the following **assumptions and constraints** [1]:

1. Each ply of the laminate is quasi-homogeneous and orthotropic.
2. The laminate is thin: its in-plane dimensions are much larger than its thickness, and it is loaded only in its own plane. As a consequence, the laminate and its plies are in a state of plane stress ($\sigma_z = \tau_{xz} = \tau_{yz} = 0$).

3. Displacements are small compared to the laminate thickness ($|u|, |v|, |w| \ll h$).
4. Displacements are continuous throughout the laminate.
5. In-plane displacements vary linearly through the thickness: u and v are linear functions of z .
6. Lines perpendicular to the mid-surface remain straight and perpendicular to that surface after deformation (**Kirchhoff-Love hypothesis**). As a consequence, transverse shear strains γ_{xz} and γ_{yz} are zero.
7. Strain-displacement and stress-strain relations are linear.
8. Normal distances from the mid-surface remain constant: the transverse normal strain ε_z is zero, which implies that the transverse displacement w is independent of coordinate z .

Hypotheses 6 and 8 structurally characterise the theory. From hypothesis 6 it follows that $\varepsilon_{xz} = \varepsilon_{yz} = 0$; from hypothesis 8, that $\varepsilon_{zz} = 0$. Figure 11 illustrates the kinematic consequences of these hypotheses on the cross-section of the plate.

Note that the normal rotation $\alpha_x = \partial w_0 / \partial x$ actually corresponds to $\tan(\alpha_x)$; under the small rotation hypothesis $\tan(\alpha_x) \approx \alpha_x$, which justifies directly identifying the slope of the mid-surface with the rotation angle of the cross-section.

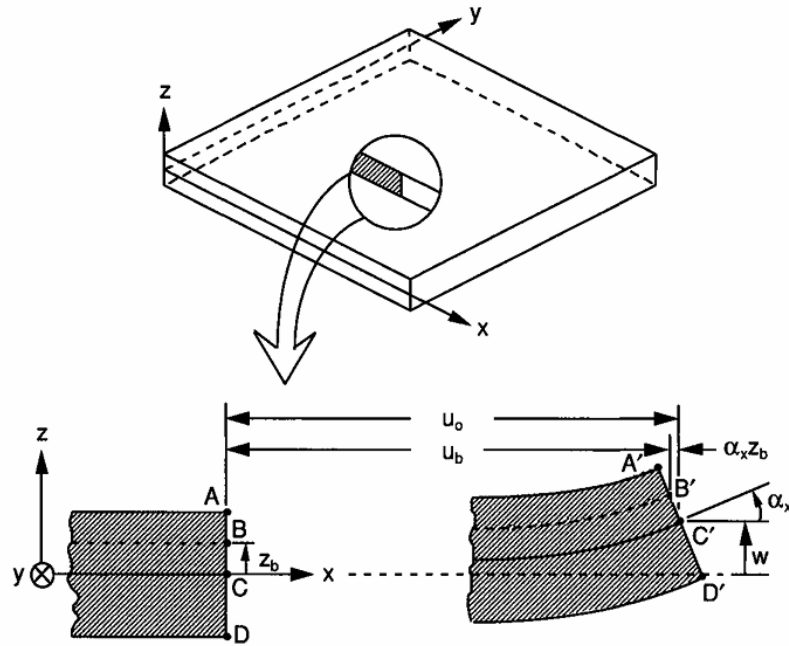


Figure 11: Kinematics of deformation of a laminate under Kirchhoff-Love hypotheses. A point B located at a distance z_b from the mid-plane undergoes a displacement $u_b = u_0 - \alpha_x z_b$, where $\alpha_x = \partial w / \partial x$ is the rotation of the normal [2].

As a consequence of the above hypotheses, the components of the displacement vector of a generic point at coordinate z are:

$$u(x, y, z, t) = u_0(x, y, t) - z \frac{\partial w_0}{\partial x}, \quad v(x, y, z, t) = v_0(x, y, t) - z \frac{\partial w_0}{\partial y}, \quad w = w_0(x, y, t) \quad (37)$$

where u_0 , v_0 and w_0 are the displacements of the mid-plane ($z = 0$).

Green-Lagrange strains in the CLT framework

Applying the displacement field (37) to the general expressions of the Green-Lagrange tensor (35), the six plate strain components are obtained:

$$\begin{aligned} E_{xx} &= \frac{\partial u}{\partial x} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 + \left(\frac{\partial w}{\partial x} \right)^2 \right] \\ E_{yy} &= \frac{\partial v}{\partial y} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial w}{\partial y} \right)^2 \right] \\ E_{zz} &= \frac{\partial w}{\partial z} + \frac{1}{2} \left[\left(\frac{\partial u}{\partial z} \right)^2 + \left(\frac{\partial v}{\partial z} \right)^2 + \left(\frac{\partial w}{\partial z} \right)^2 \right] \\ E_{xy} &= \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right) \\ E_{xz} &= \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} + \frac{\partial u}{\partial x} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial z} \right) \\ E_{yz} &= \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} + \frac{\partial u}{\partial y} \frac{\partial u}{\partial z} + \frac{\partial v}{\partial y} \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \frac{\partial w}{\partial z} \right) \end{aligned} \quad (38)$$

To simplify these expressions, the **order of magnitude** of the displacement gradients is introduced. It is assumed that in-plane gradients are of order ϵ :

$$\frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial x}, \frac{\partial v}{\partial y}, \frac{\partial w}{\partial z} = O(\epsilon) \quad (39)$$

Under the **small strain** hypothesis, terms of order ϵ^2 are negligible. These include, among others:

$$\left(\frac{\partial u}{\partial x} \right)^2, \left(\frac{\partial u}{\partial y} \right)^2, \left(\frac{\partial v}{\partial x} \right)^2, \left(\frac{\partial u}{\partial x} \right) \left(\frac{\partial u}{\partial y} \right), \dots \quad (40)$$

However, if the transverse normal rotations $\partial w_0/\partial x$ and $\partial w_0/\partial y$ are **moderate** (of the order of 10° – 15°), the quadratic terms associated with w are not negligible compared to ϵ and must be retained:

$$\left(\frac{\partial w}{\partial x}\right)^2, \left(\frac{\partial w}{\partial y}\right)^2, \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \quad (41)$$

Retaining only terms of order ϵ and the quadratic terms (41), strains (38) reduce to the so-called **von Kármán strains**:

$$\begin{aligned} \varepsilon_{xx} &= \frac{\partial u}{\partial x} + \frac{1}{2} \left(\frac{\partial w}{\partial x} \right)^2, & \varepsilon_{yy} &= \frac{\partial v}{\partial y} + \frac{1}{2} \left(\frac{\partial w}{\partial y} \right)^2 \\ \varepsilon_{xy} &= \frac{1}{2} \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x} \frac{\partial w}{\partial y} \right), & \varepsilon_{xz} &= \frac{1}{2} \left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x} \right) \\ \varepsilon_{yz} &= \frac{1}{2} \left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} \right), & \varepsilon_{zz} &= \frac{\partial w}{\partial z} \end{aligned} \quad (42)$$

Substituting now the displacement field (37) into the von Kármán strains (42), and noting that $\partial w / \partial z = 0$ in CLT, the transverse components vanish, $\varepsilon_{xz} = \varepsilon_{yz} = \varepsilon_{zz} = 0$, and the three in-plane components take the form:

$$\varepsilon_{xx} = \frac{\partial u_0}{\partial x} + \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2 - z \frac{\partial^2 w_0}{\partial x^2}, \quad \varepsilon_{yy} = \frac{\partial v_0}{\partial y} + \frac{1}{2} \left(\frac{\partial w_0}{\partial y} \right)^2 - z \frac{\partial^2 w_0}{\partial y^2} \quad (43)$$

$$\gamma_{xy} = 2\varepsilon_{xy} = \frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} + \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial y} - 2z \frac{\partial^2 w_0}{\partial x \partial y} \quad (44)$$

These are the **von Kármán strains** for the laminated plate, which account for the geometric nonlinearity due to moderate rotations. In the more restrictive case of **small deformations and small rotations**, the quadratic terms in w_0 are also neglected, so the strains linearise and take the compact matrix form:

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_s \end{bmatrix} = \underbrace{\begin{bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_s^0 \end{bmatrix}}_{\text{mid-plane}} + z \underbrace{\begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_s \end{bmatrix}}_{\text{curvatures}} \quad (45)$$

where the mid-plane membrane strains are $\varepsilon_x^0 = \frac{\partial u_0}{\partial x} + \frac{1}{2} \left(\frac{\partial w_0}{\partial x} \right)^2$, $\varepsilon_y^0 = \frac{\partial v_0}{\partial y} + \frac{1}{2} \left(\frac{\partial w_0}{\partial y} \right)^2$, $\gamma_s^0 = \frac{\partial u_0}{\partial y} + \frac{\partial v_0}{\partial x} + \frac{\partial w_0}{\partial x} \frac{\partial w_0}{\partial y}$, and the curvatures are defined as $\kappa_x = -\frac{\partial^2 w_0}{\partial x^2}$, $\kappa_y = -\frac{\partial^2 w_0}{\partial y^2}$, $\kappa_s = -2 \frac{\partial^2 w_0}{\partial x \partial y}$.

ABD matrix

Integrating stresses through the thickness of the laminate composed of n plies, the force and moment resultants per unit width are obtained.

$$\begin{bmatrix} N_x \\ N_y \\ N_s \end{bmatrix} = \begin{bmatrix} A_{xx} & A_{xy} & A_{xs} \\ A_{yx} & A_{yy} & A_{ys} \\ A_{sx} & A_{sy} & A_{ss} \end{bmatrix} \begin{bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_s^o \end{bmatrix} + \begin{bmatrix} B_{xx} & B_{xy} & B_{xs} \\ B_{yx} & B_{yy} & B_{ys} \\ B_{sx} & B_{sy} & B_{ss} \end{bmatrix} \begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_s \end{bmatrix} \quad (46)$$

$$\begin{bmatrix} M_x \\ M_y \\ M_s \end{bmatrix} = \begin{bmatrix} B_{xx} & B_{xy} & B_{xs} \\ B_{yx} & B_{yy} & B_{ys} \\ B_{sx} & B_{sy} & B_{ss} \end{bmatrix} \begin{bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_s^o \end{bmatrix} + \begin{bmatrix} D_{xx} & D_{xy} & D_{xs} \\ D_{yx} & D_{yy} & D_{ys} \\ D_{sx} & D_{sy} & D_{ss} \end{bmatrix} \begin{bmatrix} \kappa_x \\ \kappa_y \\ \kappa_s \end{bmatrix} \quad (47)$$

Combining these into a single matrix expression:

$$\begin{bmatrix} N_x \\ N_y \\ N_s \\ M_x \\ M_y \\ M_s \end{bmatrix} = \begin{bmatrix} A_{xx} & A_{xy} & A_{xs} & B_{xx} & B_{xy} & B_{xs} \\ A_{yx} & A_{yy} & A_{ys} & B_{yx} & B_{yy} & B_{ys} \\ A_{sx} & A_{sy} & A_{ss} & B_{sx} & B_{sy} & B_{ss} \\ B_{xx} & B_{xy} & B_{xs} & D_{xx} & D_{xy} & D_{xs} \\ B_{yx} & B_{yy} & B_{ys} & D_{yx} & D_{yy} & D_{ys} \\ B_{sx} & B_{sy} & B_{ss} & D_{sx} & D_{sy} & D_{ss} \end{bmatrix} \begin{bmatrix} \varepsilon_x^o \\ \varepsilon_y^o \\ \gamma_s^o \\ \kappa_x \\ \kappa_y \\ \kappa_s \end{bmatrix} \quad (48)$$

The sub-matrices $[A]$, $[B]$ and $[D]$ (3×3) are computed by summing the contribution of each ply k between its coordinates z_{k-1} (bottom face) and z_k (top face):

$$A_{ij} = \sum_{k=1}^n \bar{Q}_{ij}^{(k)} (z_k - z_{k-1}), \quad (49)$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^n \bar{Q}_{ij}^{(k)} (z_k^2 - z_{k-1}^2), \quad (50)$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^n \bar{Q}_{ij}^{(k)} (z_k^3 - z_{k-1}^3). \quad (51)$$

where $i, j = x, y, s$.

The sub-matrix $[A]$ is the **extensional stiffness**, $[D]$ the **bending stiffness**, and $[B]$ the **coupling stiffness** that couples extension with bending. For laminates symmetric about the mid-plane, $[B] = [0]$.

Effective elastic properties of the laminate

To obtain the effective engineering constants of the laminate of thickness h , the sub-matrix $[a] = [K]^{-1}$ is used, where $[K] = [A] - [B][D]^{-1}[B]$. For symmetric laminates ($[B] = [0]$), $[K] = [A]$. The effective properties are:

$$\begin{aligned}\bar{E}_x &= \frac{1}{h a_{xx}}, & \bar{E}_y &= \frac{1}{h a_{yy}}, & \bar{G}_{xy} &= \frac{1}{h a_{ss}}, \\ \bar{\nu}_{xy} &= -\frac{a_{yx}}{a_{xx}}, & \bar{\nu}_{yx} &= -\frac{a_{xy}}{a_{yy}}.\end{aligned}\quad (52)$$

```

1 % Laminate definition: [angle, E1, E2, G12, nu12, thickness, F1t, F1c,
  F2t, F2c, F6]
2 M1 = [225e3, 6.68e3, 4.08e3, 0.32, 0.124, 1816, 854, 31.4, 235, 75.4];
3 M2 = [73.9e3, 71.5e3, 4.03e3, 0.07, 0.209, 1106, 969, 776, 738, 117];
4
5 laminate = [0, M2;
6             0, M1;
7             45, M2];
8
9 % --- Total thickness and z0 ---
10 thk0 = 0;
11 for i = 1:size(laminate,1)
12     thk0 = thk0 + laminate(i,6);
13 end
14 h = thk0;
15 z0 = -h/2;
16
17 % --- Loop over each ply ---
18 A0 = zeros(3); B0 = zeros(3); D0 = zeros(3);
19 zkm1 = z0;
20
21 for i = 1:size(laminate,1)
22     theta = laminate(i,1);
23     E1 = laminate(i,2); E2 = laminate(i,3);
24     G12 = laminate(i,4); nu12 = laminate(i,5);
25     nu21 = E2*nu12/E1;
26     zk = zkm1 + laminate(i,6);
27     m = cosd(theta); n = sind(theta);
28
29     Q11 = E1/(1-nu12*nu21); Q22 = E2/(1-nu12*nu21);
30     Q12 = nu21*E1/(1-nu12*nu21); Q66 = G12;
31
32     % Transformed Q_bar components
33     Qxx = m^4*Q11+n^4*Q22+2*m^2*n^2*Q12+4*m^2*n^2*Q66;
34     Qyy = n^4*Q11+m^4*Q22+2*m^2*n^2*Q12+4*m^2*n^2*Q66;
35     Qxy = m^2*n^2*(Q11+Q22)+(m^4+n^4)*Q12-4*m^2*n^2*Q66;
36     Qxs = m^3*n*Q11-m*n^3*Q22-m*n*(m^2-n^2)*(Q12+2*Q66);
37     Qys = m*n^3*Q11-m^3*n*Q22+m*n*(m^2-n^2)*(Q12+2*Q66);
38     Qss = m^2*n^2*(Q11+Q22-2*Q12)+(m^2-n^2)^2*Q66;
39
40     Qbar = [Qxx Qxy Qxs; Qxy Qyy Qys; Qxs Qys Qss];
41
42     A0 = A0 + Qbar*(zk-zkm1);
43     B0 = B0 + Qbar*(zk^2-zkm1^2)/2;
44     D0 = D0 + Qbar*(zk^3-zkm1^3)/3;
45     zkm1 = zk;
46 end
47 A = A0; B = B0; D = D0;

```

```

48
49 % For symmetric laminates, B = 0
50 if all(abs(B(:)) < 1e-10), B = zeros(3); end
51
52 % Effective properties
53 K = A - B*(D\B);
54 a = K \ eye(3);
55 Ex = 1/(h*a(1,1))
56 Ey = 1/(h*a(2,2))
57 Gxy = 1/(h*a(3,3))
58 poissonxy = -a(2,1)/a(1,1)

```

Listing 4: Construction of the ABD matrix and effective laminate properties

4.6 Strength of a multidirectional laminate

Stress distribution through the thickness

For a given force and moment state $\{N_x, N_y, N_s, M_x, M_y, M_s\}$, mid-plane strains and curvatures are obtained by inverting the system (48):

$$\begin{bmatrix} \epsilon^0 \\ \kappa \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix}^{-1} \begin{bmatrix} N_x \\ N_y \\ N_s \\ M_x \\ M_y \\ M_s \end{bmatrix}. \quad (53)$$

For ply k , the strains at its bottom face ($z = z_{k-1}$) and top face ($z = z_k$) in the laminate axes are computed applying Equation (45):

$$\epsilon_{xy}^{\text{bot/top}} = \epsilon^0 + z_{k-1/k} \kappa. \quad (54)$$

To transform strains to the ply's principal axes (1, 2), matrix $[T]$ is applied including the $\frac{1}{2}$ factor on the engineering shear term:

$$\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \frac{1}{2}\gamma_6 \end{bmatrix} = [T] \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \frac{1}{2}\gamma_s \end{bmatrix}. \quad (55)$$

Stresses in principal axes are obtained by multiplying by the ply's $[Q]$ matrix (restoring the factor 2 on the shear term):

$$\begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_6 \end{bmatrix} = [Q] \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_6 \end{bmatrix}. \quad (56)$$

Application of the Tsai-Wu criterion ply by ply

Once the stresses $(\sigma_1, \sigma_2, \tau_6)$ at the top and bottom faces of each ply are known, the Tsai-Wu criterion (Equation (25)) is applied with the strength coefficients of the corresponding ply, and

the safety factor is calculated using (28). The **global laminate safety factor** is the minimum over all plies and all evaluated z positions.

```

1  % Applied forces and moments (per unit width)
2  Nx = 1; Ny = 0; Ns = 0;
3  Mx = 0; My = 0; Ms = 0;
4
5  midstrains_and_curvatures = [A, B; B, D] \ [Nx;Ny;Ns;Mx;My;Ms];
6
7  zkm1 = z0;
8  Sfa = zeros(size(laminate,1), 2);
9  Sfr = zeros(size(laminate,1), 2);
10
11 for i = 1:size(laminate,1)
12     theta = laminate(i,1);
13     E1 = laminate(i,2); E2 = laminate(i,3);
14     G12 = laminate(i,4); nu12 = laminate(i,5);
15     nu21 = E2*nu12/E1;
16     zk = zkm1 + laminate(i,6);
17     m = cosd(theta); n = sind(theta);
18
19     Q11=E1/(1-nu12*nu21); Q22=E2/(1-nu12*nu21);
20     Q12=nu21*E1/(1-nu12*nu21); Q66=G12;
21     Q=[Q11 Q12 0; Q12 Q22 0; 0 0 Q66];
22     T=[m^2 n^2 2*m*n; n^2 m^2 -2*m*n; -m*n m*n m^2-n^2];
23     R=[1 0 0; 0 1 0; 0 0 0.5]; % inverse Reuter matrix
24
25     for side = 1:2
26         if side==1, zs=zkm1; else, zs=zk; end
27         eps_xy = midstrains_and_curvatures(1:3) + zs*
28             midstrains_and_curvatures(4:6);
29         eps_12 = T*(R*eps_xy); % [e1; e2; 0.5*g6]
30         sig_12 = Q*([1 0 0; 0 1 0; 0 0 2]*eps_12); % [s1; s2; t6]
31
32         sigma1=sig_12(1); sigma2=sig_12(2); tau6=sig_12(3);
33
34         % Tsai-Wu coefficients
35         F1t=laminate(i,7); F1c=laminate(i,8);
36         F2t=laminate(i,9); F2c=laminate(i,10); F6=laminate(i,11);
37         f1=1/F1t-1/F1c; f2=1/F2t-1/F2c;
38         f11=1/(F1t*F1c); f22=1/(F2t*F2c); f66=1/F6^2;
39         f12=-0.5*sqrt(f11*f22);
40
41         a_tw = f11*sigma1^2 + f22*sigma2^2 + f66*tau6^2 + 2*f12*sigma1*
42             sigma2;
43         b_tw = f1*sigma1 + f2*sigma2;
44         disc = sqrt(b_tw^2 + 4*a_tw);
45
46         Sfa(i,side) = (-b_tw + disc)/(2*a_tw);
47         Sfr(i,side) = abs((-b_tw - disc)/(2*a_tw));
48     end
49     zkm1 = zk;
50 end
51 Sfa_min = min(Sfa, [], "all") % Global forward safety factor

```

```
51 Sfr_min = min(Sfr, [], "all") % Global reverse safety factor
```

Listing 5: Evaluation of the Tsai-Wu criterion and global safety factor

The result for the laminate $[0_{M2}/0_{M1}/45_{M2}]$ subjected to $N_x = 1 \text{ N/mm}$ is $SF_a = 283.4$ and $SF_r = 103.0$, indicating that the critical ply is the 45° ply subjected to in-plane shear.

4.7 Analysis of laminated beams: three-point bending test

Plate-to-beam reduction

When the panel span a is much larger than its width b , the transverse variation of deflection w_0 in the y direction can be neglected and the three-dimensional problem reduces to a one-dimensional one analogous to Euler-Bernoulli beam theory. For a laminate without in-plane forces, the curvature equations follow from relation (47). Additionally, for a symmetric laminate ($[B] = [0]$), this reduces to:

$$\begin{bmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{bmatrix} = - \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} \partial^2 w_0 / \partial x^2 \\ \partial^2 w_0 / \partial y^2 \\ 2 \partial^2 w_0 / \partial x \partial y \end{bmatrix} \quad (57)$$

For the one-dimensional analysis, $M_{yy} = M_{xy} = 0$ is imposed, leading to the relations:

$$\frac{\partial^2 w_0}{\partial x^2} = -D_{11}^* M_{xx}, \quad \frac{\partial^2 w_0}{\partial y^2} = -D_{12}^* M_{xx}, \quad 2 \frac{\partial^2 w_0}{\partial x \partial y} = -D_{16}^* M_{xx} \quad (58)$$

where D_{ij}^* are the elements of the flexural compliance matrix $[\mathbf{D}]^{-1}$. The first equation governs the deflection in x and is directly analogous to the Euler-Bernoulli beam equation, with effective stiffness $EI \equiv b/D_{11}^*$. The other two equations describe secondary effects: the second corresponds to the anticlastic bending due to the Poisson effect (D_{12}^*), and the third to a bending-torsion coupling (D_{16}^*) that only exists in laminates where $D_{16} \neq 0$ (angle-ply laminates).

For the beam approximation to be valid, the variation of deflection in y must be negligible compared to that in x . This requires $|D_{16}^*/D_{11}^*| \ll 1$ or, equivalently, that the ratio a/b be sufficiently large. In cross-ply laminates ($[0/90]_s$) the coefficient $D_{16} = 0$ exactly, so the condition is satisfied at smaller a/b ratios than for angle-ply laminates.

Deflection under central point load

Consider a simply supported laminated beam of span a and width b , subjected to a total load F applied at the centre ($x = a/2$). The bending moment distribution per unit width in the left half-span ($0 \leq x \leq a/2$) is:

$$M_{xx}(x) = \frac{F}{2b} x = f_0 \frac{x}{2}, \quad f_0 = \frac{F}{b} \quad [\text{N/mm}] \quad (59)$$

Substituting into the first equation of (58) and integrating twice, with boundary conditions $w_0(0) = 0$ and $dw_0/dx|_{x=a/2} = 0$ (symmetry at the centre), the deflection profile is obtained:

$$w_0(x) = \frac{D_{11}^* f_0}{48} (3a^2x - 4x^3), \quad 0 \leq x \leq \frac{a}{2} \quad (60)$$

The maximum deflection occurs at the centre, $x = a/2$:

$$w_{\max} = \frac{D_{11}^* F a^3}{48 b} \quad (61)$$

This expression is the laminated equivalent of the classical Euler-Bernoulli formula $w_{\max} = FL^3/(48EI)$, with $D_{11}^* = 1/(EI/b)$ for a homogeneous cross-section.

Validity of the model: small deflections

Equation (61) is based on the linearised curvature $\kappa \approx d^2w_0/dx^2$, which approximates the exact curvature:

$$\kappa = \frac{d^2w_0/dx^2}{[1 + (dw_0/dx)^2]^{3/2}} \quad (62)$$

The approximation is valid while $(dw_0/dx)^2 \ll 1$. As a practical rule, small deflection theory is accurate when $w_{\max}/a < 5\%$. For larger deflections, the real curvature is always *less* than the linearised one for the same geometric deflection; the solid must deform more to maintain equilibrium with the applied load, so CLT *overestimates* the stiffness and *underestimates* the real deflection.

MATLAB implementation

```

1 % --- Test parameters ---
2 a = 180; % span (mm)
3 b = 40; % width (mm)
4 F_max = 200; % maximum applied force (N)
5
6 % --- Bending stiffness: D11* = inv(D)(1,1) ---
7 % D is the 3x3 bending sub-matrix built in the ABD loop above
8 Dstar = inv(D); % flexural compliance matrix
9 D11s = Dstar(1,1); % (mm / N*mm) = 1/(N*mm/mm)
10
11 % --- Theoretical maximum deflection ---
12 % w_max = D11* * (F/b) * a^3 / 48
13 w_max_por_N = D11s * a^3 / (48 * b); % mm/N
14 fprintf('Theoretical stiffness: k = %.4f N/mm\n', 1/w_max_por_N);
15
16 % --- Force-deflection curve (x = a/2) ---
17 F_vec = linspace(0, F_max, 500);
18 w_vec = w_max_por_N * F_vec; % mm
19
20 % --- Deflection profile for F = F_max ---

```

```

21 x = linspace(0, a/2, 300);
22 f0 = F_max / b; % N/mm (load per unit width)
23 w_perfil = D11s * f0 / 48 .* (3*a^2*x - 4*x.^3);
24
25 % --- Small deflection limit (w/a = 5%) ---
26 w5pct = 0.05 * a;
27 fprintf('Small deflection limit: w_max/a = 5% -> w = %.1f mm\n', w5pct)
28 ;
29 fprintf('Theoretical max deflection at F = %.0f N: w_max = %.2f mm\n',
    F_max, w_max_por_N*F_max);
30 fprintf('Ratio w_max/a = %.1f%%\n', w_max_por_N*F_max/a*100);

```

Listing 6: Analysis of a simply supported laminated beam under central point load (3PBT)

Experimental validation

To verify the analytical model, three-point bending tests were carried out on three CFRP laminate configurations, all with span $a = 180$ mm, width $b = 40$ mm and 8 carbon/epoxy plies of $t = 0.209$ mm each. The configurations tested were:

- **Test 1 — T0** $[0^\circ]_8$: unidirectional woven laminate at zero degrees.
- **Test 2 — T45** $[\pm 45^\circ]_{4s}$: woven laminate at $\pm 45^\circ$, symmetric.
- **Test 3 — UD0** $[0^\circ]_8$: unidirectional (UD) laminate at zero degrees.

The results are summarised in Table 1 and Figure 12.

Table 1: Summary of three-point bending tests. The modulus E_{auto} is the value automatically reported by the testing machine in the linear regime.

Test	Configuration	F_{max} (N)	w_{max} (mm)	w_{max}/a (%)	E_{auto} (MPa)
1	T0 $[0^\circ]_8$	200	25.84	14.4	66 684
2	T45 $[\pm 45^\circ]_{4s}$	50	26.48	14.7	14 092
3	UD0 $[0^\circ]_8$	150	34.47	19.1	42 547

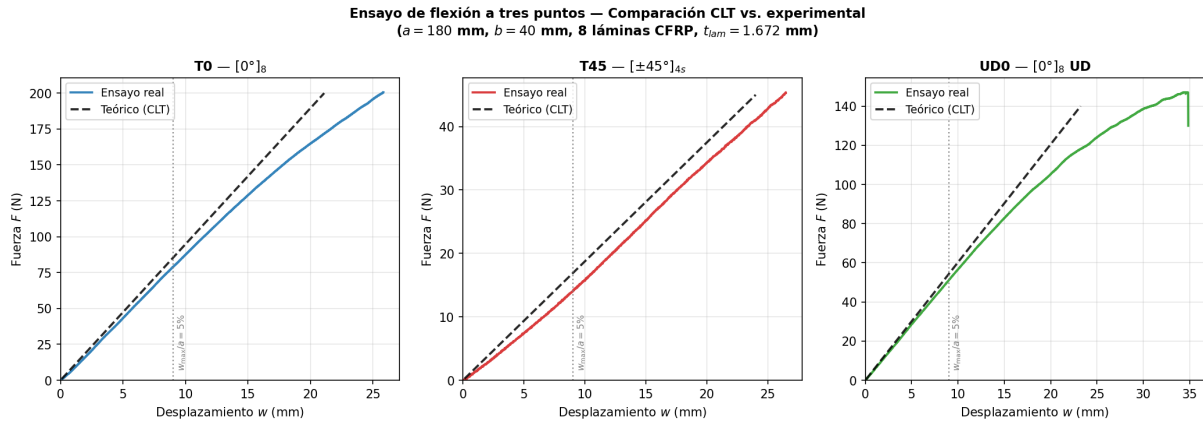


Figure 12: Comparison between the experimental force-deflection curve and the CLT theoretical prediction (Equation (61)) for the three configurations tested. The vertical dotted line indicates the $w_{max}/a = 5\%$ limit beyond which the small deflection hypothesis is no longer strictly valid.

In all three tests, the experimental and theoretical curves agree well in the small deflection regime ($w < 9$ mm, i.e. $w/a < 5\%$), with stiffness differences below 6% for Test T0. As deflection increases, the experimental curve departs from the linear prediction: the real panel is more flexible than CLT predicts, a behaviour explained by **geometric nonlinearity**.

The physical mechanism is as follows: the real curvature (62) is always less than the linearised curvature d^2w_0/dx^2 because the denominator $(1 + (dw_0/dx)^2)^{3/2} > 1$. For the same geometric deflection, the real beam generates less internal bending moment than linear theory predicts. To maintain equilibrium with the applied load, it must deform more, resulting in an apparently lower stiffness. For the tests conducted, with w_{max}/a between 14% and 19%, the geometric correction is significant and the linear model appreciably overestimates the stiffness [2].

Test UD0 (Test 3) also shows a marked nonlinear behaviour and a subsequent load drop due to cross-section failure, visible as the final descent in the experimental curve.

5 Budget

6 Environmental impact

Conclusions

Acknowledgements

Thanks.

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